



Optimized ANN-based mathematical model and software application for predicting raw mix lime saturation factor for high-quality cement production

Lateef Bankole Adamolekun ^{a,b,*} 

^a Mining Engineering Department, Dangote Cement, Gboko, Benue State, Nigeria

^b Mining Engineering Department, School of Engineering and Engineering Technology, Federal University of Technology, Akure, Nigeria

ARTICLE INFO

Editor: DR B Gyampoh.

Keywords:

Cement raw material
Lime saturation factor
Chemical composition
Levenberg-Marquardt-Artificial neural network
Particle swarm optimization-Artificial neural network
Genetic algorithms-Artificial neural network

ABSTRACT

The rapid increase in population drives a growing demand for expanded infrastructure. Cement quality, crucial for infrastructure performance, is largely influenced by a well-controlled raw mix lime saturation factor (LSF). Accurate LSF estimation relies on integrating precise mathematical formulas into elemental composition analyzers. However, the formulas traditionally utilized in the cement industry, often fall short of capturing underlying complexities of the process. Thus, there is need for more robust mathematical formula to accurately estimate LSF. This study develops LSF predictive models by employing artificial neural networks (ANN) optimized with particle swarm optimization (PSO), Levenberg-Marquardt (LM), and genetic algorithms (GA), using two thousand four hundred and sixty data points obtained via cross belt-analyzer. Dependable variables selected were lime, silica, alumina, and iron oxide. To enhance the practicality and ease of use, the models (LM-ANN, PSO-ANN, and GA-ANN) were converted into mathematical equations and further integrated into software application, in form of simple calculator. The models were validated using 5-fold cross-validation with random sampling, demonstrating consistent, generalization capability, and reliable performance across key metrics including coefficient of determination (R^2), root mean squared error (RMSE), and mean absolute error (MAE). The models' performance was benchmarked against the established model proposed by Bogue (1966). The LM-ANN model outperformed both Bogue's and the other evaluated models, achieving superior results across key metrics: $R^2 = 0.9885$, RMSE = 1.7828, relative squared error (RSE) = 9.99×10^{-7} . While all three models are suitable for practical deployment, the LM-ANN model is strongly recommended for industrial applications. The mathematical model can be integrated into elemental composition analyzers to enhance real-time process optimization

Abbreviations: LSF, lime saturation factor; LM-ANN, Levenberg-Marquardt- artificial neural network; GA-ANN, genetic algorithms-artificial neural network; PSO-ANN, particle swarm optimization-artificial neural network; LSF_{LM-ANN} , lime saturation factor predicted by Levenberg-Marquardt-artificial neural network; $LSF_{PSO-ANN}$, lime saturation factor predicted by particle swarm optimization-artificial neural network; LSF_{GA-ANN} , lime saturation factor predicted by genetic algorithms-artificial neural network; R, coefficient of correlation; R^2 , coefficient of determination; MSE, mean squared error; RMSE, root mean squared error; MAE, mean absolute error; MAPE, mean absolute percentage error; RSE, relative squared error; CaO, lime; SiO₂, silica; Al₂O₃, alumina; Fe₂O₃, iron oxide; CBA, cross-belt analyzers; XRF, X-ray Fluorescence; SVR, support vector regression; LS-SVM, least squares support vector machines; MVR, multivariate regression; HPBR-ANN, hunter point back propagation-artificial neural network; PGNA, prompt gamma neutron activation analysis; RBF, radial basis function.

* Corresponding author at: Mining Engineering Department, School of Engineering and Engineering Technology, Federal University of Technology, Akure, Nigeria.

E-mail address: adamolekunbankole@gmail.com.

<https://doi.org/10.1016/j.sciaf.2026.e03195>

Received 20 April 2025; Received in revised form 22 December 2025; Accepted 26 January 2026

Available online 27 January 2026

2468-2276/© 2026 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

and improve cement production efficiency. Meanwhile, the software application will serve as a smart tool for rapid LSF estimation and consistent monitoring of analyzer reliability in cement production.

Introduction

The swift rise in population is driving a heightened demand for expanded infrastructure. Cement serves as a foundational material in contemporary development, enabling the creation of roads, bridges, buildings, water systems, and other vital undertakings that fuel economic development. Cement is a fine powdered substance made from clinker, gypsum, and other additives grounded together, which develops strong adhesive and cohesive properties when mixed with water [1]. It is a binding agent used in construction to hold together aggregates like sand, gravel, and crushed stone. Cement product quality is largely determined by clinker quality [1]. Clinker is the main component of cement; made up of about 70–80 % of the total weight of cement [2]. It is a nodular material with a diameter of 3–25 mm, formed by the reaction of limestone, clay and other minerals inside the kiln, at high temperature, around 1450°C [3]. Carbonate rocks such as limestone are very important in the manufacture of clinker; when they are heated in the kiln, they decompose into lime, which provides the needed binding agent for mixing with other materials to form clinker for cement production [4].

However, all the lime in the clinker raw mix may not react with silica, alumina and iron oxide during the clinker production process. The lime content need not be excessively high to prevent the formation of free lime, which negatively impacts cement quality [2]. The leftover lime remains in its free form reacts with water over time, leading to expansion and cracking in the hardened cement. Presence of free lime can also lead to issues like poor grindability, increase setting time, reduced strength, and decreased durability [4]. For instant, in Sanusi [1], the author collected 10g samples of clinker, and were milled and pelletized with the aid of pyridine and binding agent. The pelletized samples were subjected to grinding for 40 min using ball mill after 5 % gypsum. He concluded that the energy required in grinding the clinker increase with increase in the free lime. Meanwhile, Sanusi et al. [5] conducted research to determine the effect and control of free lime of raw materials for clinker production and the authors discovered that the burnability of the clinker increases with increase in free lime. On the other hand, lime content of raw mix need not to be too low for cement production. When there is insufficient lime to react with the oxides, key compounds like tricalcium silicate (C_3S) that is responsible for early strength won't form sufficiently and this will results in lower compressive strength of the cement [4,3]. Thus, there is need for proper control to minimize the presence of free lime and ensure good cement quality. Sanusi [1] and Sanusi et al. [5] concluded that to minimize the free lime, there is needs to ensure the implementation of control measures of lime saturation factor (LSF). The LSF is a crucial parameter in cement chemistry, used to determine whether there is an optimal balance of lime in the raw materials for producing cement clinker [6]. Specifically, it helps ensure there is enough lime to combine with silica, alumina, and iron oxide to form the desired cement compounds, without leaving excess free CaO . If LSF is too high, it indicates that there is too much free lime. On the other hand, when LSF is too low, it shows that there is insufficient lime to react with the oxides [4,3]. By controlling the LSF within an optimal range, cement manufactures can minimize free lime and produce high quality cement [7]. Thus, there is need for continues LSF estimation based on the chemical composition of the cement raw material for close monitoring.

Cross-belt analyzers (CBA) and X-ray fluorescence (XRF) are the most commonly used instruments in the cement industry for estimating the LSF to ensure proper control. CBA is an online analyzer that measures the elemental concentration of raw materials as they move along a conveyor belt using prompt gamma neutron activation analysis (PGNAA) or other nuclear techniques. It provides real-time analysis of raw mix composition, enabling quick adjustments to the LSF. CBA ensures accurate and reliable measurements of elemental compositions in the raw mix of limestone, which are then used to estimate the LSF. X-ray Fluorescence (XRF), on the other hand, analyzes the elemental composition of a sample by detecting secondary X-rays emitted when the sample is exposed to high-energy X-rays. While XRF results are not instantaneous due to the time required for sample preparation and analysis, this method is highly accurate for determining the concentrations of lime (CaO), silica (SiO_2), alumina (Al_2O_3), and iron oxide (Fe_2O_3), which are critical components for LSF calculation. However, CBA and XRF rely on pre-programmed formulas to calculate LSF based on the chemical composition (CaO , SiO_2 , Al_2O_3 , and Fe_2O_3) measured.

Bogue [8] proposed a mathematical formula that relates the chemical composition of cement raw material to the LSF. This formula is a well-established and widely adopted as industrial standard in the cement industry. However, Bogue's mathematical formula is a simplified relationships between variables based on observation, experience, and intuition. As a result, they may not fully capture the complexity of the underlying mechanisms. This limitation can lead to reduced accuracy, particularly when extrapolating to new conditions. In contrast, machine learning techniques have been recognized as a superior alternative to empirical models [9].

Machine learning (ML) is an artificial intelligent (AI) application that allows systems to learn from their environment automatically and improve by the experience they gathered from the learning [10,11]. There are different numbers of common algorithms used in the ML, this include neural networks, support vector machines, decision trees, random forest, logistic regression, and many more. ML has been used to develop different models to solve different science and engineering problem in cement industry today. For example, Wu, et al. [12] developed fuzzy neural network with particle swarm optimization (FNN-PSO) methods and artificial neural network (ANN) to establish and optimize cement raw material blending process. The authors concluded that the models performed excellently. Yuan et al. [13] developed models to predict free lime in clinker using artificial neural networks (ANN) with radial basis function (RBF) activation. The authors reported excellent predictive performance, demonstrating the model's practicality. Tiryaki et al. [7] developed a mathematical model for the cement raw-material blending process using neural networks. The study utilized both linear and nonlinear neural network models to simulate the blending process. The models' predictions were then compared with experimental

data. The results indicated that the models performed effectively, with the nonlinear model demonstrating higher predictive accuracy. Similarly, Zhao et al. [14] developed a model employing the particle swarm optimization (PSO) algorithm to predict free lime content, which also performed remarkably well.

In the realm of lime saturation factor (LSF), a critical determinant of cement quality, recent researchers like Taiwo et al. [2] have proposed mathematically driven AI models for predicting rock lime saturation factors in cement production using carbon, hydrogen, and oxygen as dependent variables. Their study utilized one hundred and forty four (144) data points to develop models incorporating techniques such as multilayer perceptrons (MLP), least squares support vector machines (LS-SVM), multivariate regression (MVR), and the hunter point back propagation-artificial neural network (HPBR-ANN). The authors concluded that the models performed exceptionally well and were suitable for practical applications. However, the input parameters in Taiwo et al. [2] study differ from those in the current research, which are CaO , SiO_2 , Al_2O_3 , and Fe_2O_3 . Furthermore, Kermani and Behbahani [15] utilized artificial neural networks to develop a model for predicting LSF from the chemical composition (CaO , SiO_2 , Al_2O_3 , and Fe_2O_3) in cement raw mix. The authors concluded the model presented outstanding results, making it suitable for practical purposes. Similarly, Zhang and Li [16] employed support vector regression (SVR) to predict LSF based on chemical composition. The authors concluded the model was practically applicable. However, a notable limitation of the studies that modeled LSF from the chemical composition is that their models were not translated into mathematical equations or implemented in software applications, which may hinder ease of use for end-users.

Among the machine learning techniques, artificial neural networks has been found to be more flexible [17], leading to its high usage among researchers [15,9,2]. Another important quality of ANN that made it more popular among researcher nowadays is its ability to be converted from black box to white box, that is, it can be converted to mathematic model for users' easy application. This made it a favorite for this research, which aimed at developing mathematical model for predicting cement raw mix LSF from the chemical composition. However, one of the disadvantages of ANNs is that it can easily overfit to the training data, especially when the network is too complex, and this can lead to poor generalization on unseen data [18]. The enhancement of ANNs through optimization methods can mitigate this disadvantage by identifying the best network structure using appropriate optimization techniques [19]. Thus, this research builds on previous studies by using PSO-ANN, GA-ANN, and LM-ANN algorithms to develop mathematical-based models for predicting LSF based on the chemical composition (CaO , SiO_2 , Al_2O_3 , and Fe_2O_3). Additionally, this research takes a step further by developing a software application embedding the transformed mathematical equations of the proposed predictive models. This innovation will enhance cement quality, reduces production costs, and boosts efficiency in cement manufacturing.

Methodology

This section presents the materials, procedures, and analytical methods employed to ensure accurate prediction of the LSF in limestone raw mixes, as illustrated in the flowchart shown in Fig. 1. A total of 2,560 data points were collected using CBA. Of these, 2,460 data points were used to develop predictive models, including LM-ANN, PSO-ANN, and GA-ANN. Following models development, the models were converted into explicit mathematical equations and integrated into a user-friendly software application to facilitate practical implementation. To assess their reliability and generalization on new data, the model validation was performed using a 5-fold cross-validation technique with random sampling. The performance of each model was evaluated using standard metrics such as R^2 , RMSE, and MAE. To identify the best, the developed models' performance was benchmarked against the traditional Bogue's equation.

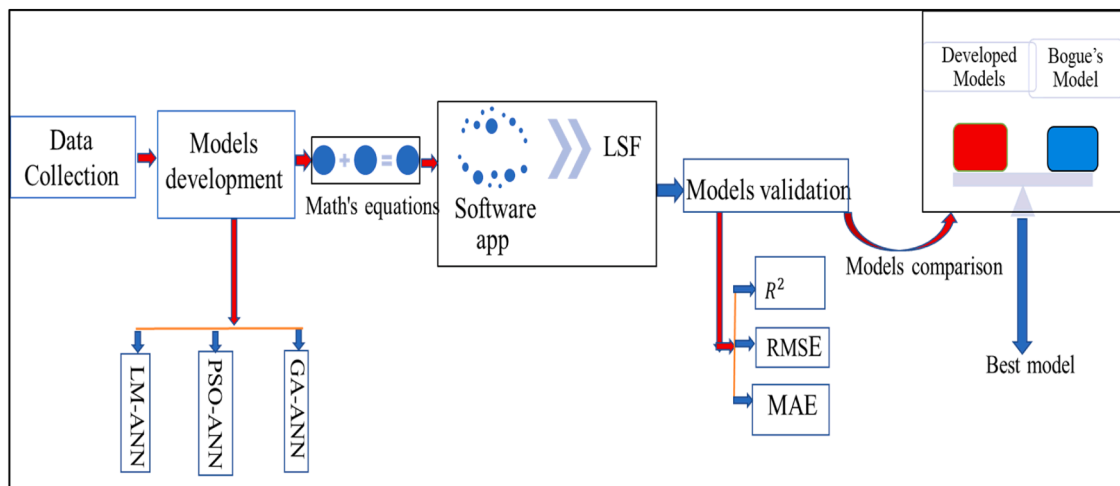


Fig. 1. Flow chart of the research design.

Data collection

This research utilizes online analytical data of limestone raw mixes obtained from Dangote Cement Plants located in Gboko (Benue State), Obajana (Kogi State), and Ibese (Ogun State), as illustrated in Fig. 2. The data, collected through CBA, comprises a total of 2,560 analyzed data points; 900 each from the Obajana and Ibese plants, and 1,040 from the Gboko plant. CBA specifically measured the concentrations of CaO , SiO_2 , Al_2O_3 , and Fe_2O_3 , while the LSF was estimated using a pre-programmed formula. To identify trends and anomalies within the dataset, a statistical analysis was performed and is presented in the subsequent subsection.

Statistical description of the dataset

Statistical description refers to the process of summarizing and explaining the key characteristics of a dataset using statistical measures such as mean, median, range, and standard deviation, among others. It aids in identifying the shape, central tendency, and variability of the data, which is crucial for selecting appropriate modeling techniques. The statistical description of the data used in this research is presented in Table 1. The composition of the data shows variability in the following ranges: SiO_2 from 2.79 % to 18.28 %, Al_2O_3 from 0.58 % to 6.77 %, Fe_2O_3 between 0.25 % and 4.73 %, and CaO ranging from 33.48 % to 49.83 %. Also, LSF values ranged widely from 55.63 to 438.57.

As highlighted in Table 1, there is notable variability in the dataset, particularly for LSF. This is evident from its high standard deviation and significantly elevated kurtosis, which indicate non-normality. As a result, the data is unsuitable for regression analysis; it cannot adequately capture the inherent variability in geomaterial properties [9]. To address this challenge, complex models are required to uncover the underlying patterns and relationships in the data. Soft computing techniques, often employed when traditional methods prove ineffective due to uncertainty, imprecision, or noise [20], and were utilized for modeling LSF based on SiO_2 , Al_2O_3 , Fe_2O_3 , and CaO in this research. To identify relationships within the dataset, a correlation test was performed, and the details are presented in the subsequent subsection.

Data parameters correlation

Correlation of data parameters refers to the statistical measure that describes the relationship between two variables within a dataset, indicating how changes in one variable affect the other. It measures both the strength and direction of the relationship



Fig. 2. Map of Nigeria showing the study location.

Table 1
Statistical description of the experimental dataset.

Statistic	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	LSF
Mean	10.52253	3.693175	1.75241	41.88425	124.0455
Standard Error	0.028901	0.013459	0.008358	0.036166	0.717854
Median	10.59	3.7	1.74	41.77	119.02
Mode	10.85	3.92	1.79	41.6	118.69
Standard Deviation	1.462281	0.680958	0.422903	1.829866	36.32086
Kurtosis	8.658145	4.410004	6.106371	2.343463	31.31899
Skewness	-1.4441	-0.91313	1.261205	0.411787	5.054572
Minimum	2.79	0.58	0.25	33.48	55.63
Maximum	18.28	6.77	4.73	49.83	438.57
Count	2560	2560	2560	2560	2560

between variables. The strength of this relationship is represented by the correlation coefficient, a numerical value ranging from -1 to 1. Values closer to ± 1 indicate a strong correlation, whereas values near zero signify a weak correlation. Correlation can either be positive or negative, depending on the nature of the relationship between the variables. A positive correlation means that an increase in one variable is associated with an increase in the other, verse versa. To analyze the correlation among limestone raw mix parameters (SiO₂, Al₂O₃, Fe₂O₃, CaO, and LSF), this research utilized Pearson's correlation matrix, as shown in Table 2. According to Dancey and Reidy [21], a Pearson's correlation coefficient of ± 1 represents a perfect correlation, ± 0.7 to 0.9 signifies a strong correlation, ± 0.4 to 0.6 indicates a moderate correlation, and ± 0.1 to 0.3 reflects a weak correlation.

The matrix (Table 2) revealed a very strong positive correlation between SiO₂ and Al₂O₃, indicating that as SiO₂ increases; Al₂O₃ also increases, and vice versa. A very strong negative correlation was observed between SiO₂ and both CaO and LSF, meaning that an increase in SiO₂ results in a decrease in CaO and LSF.

Furthermore, a moderate positive correlation was identified between SiO₂ and Fe₂O₃, suggesting that an increase in SiO₂ leads to only a slight increase in Fe₂O₃. The case of Al₂O₃ is similar to that of SiO₂; however, Al₂O₃ shows a moderate negative correlation with CaO. The correlation matrix also revealed a weak negative correlation between Fe₂O₃ and CaO, implying that changes in CaO slightly affect Fe₂O₃. Additionally, a moderate negative correlation was found between Fe₂O₃ and LSF. Lastly, a strong positive correlation was observed between CaO and LSF, indicating that an increase in CaO causes a sharp increase in LSF.

In summary, the correlation matrix shows that LSF increases with an increase in CaO, while it decreases with an increase in SiO₂, Al₂O₃, and Fe₂O₃. Accordingly, this research aims to develop an LSF predictive model using SiO₂, Al₂O₃, Fe₂O₃, and CaO as input parameters. The findings from the correlation matrix confirm that LSF is influenced by these variables, thereby validating their suitability as input parameters for improving model development and predictive performance. In this study, LSF prediction models were developed using PSO-ANN, GA-ANN, and LM-ANN architectures, with SiO₂, Al₂O₃, Fe₂O₃, and CaO serving as input parameters, as detailed in next subsection.

Model development

The proposed predictive models were developed to improve the prediction of LSF, a process traditionally requiring costly and labor-intensive methods. To create these models, the dataset was first divided using a block-splitting method into a modeling dataset with Two thousand four hundred and sixty (2,460) data points and a trial dataset with hundred (100) data points for the development and validation of the models, respectively. These predictive models aim to estimate LSF from SiO₂, Al₂O₃, Fe₂O₃, and CaO, which can be easily measured using XRF, thus bypassing the need for expensive and tedious tests. The proposed soft computing models include LM-ANN, PSO-ANN, GA-ANN. These multi-input, single-output models were trained on a dataset of 2,460 data points. The dataset was normalized to a range of -1 to 1 using Eq. (1) to ensure dimensional consistency among the parameters and to prevent over fitting during network training [20].

$$Y_i = \frac{(R_{max} - R_{min})(X_i - X_{min})}{X_{max} - X_{min}} + R_{min} \quad (1)$$

where Y_i is the normalized parameter, X_i is the actual data to be normalized, R_{max} is the maximum value of the range of normalization, R_{min} is the minimum value of the range of normalization, and X_{max} and X_{min} are the maximum and minimum values of the actual data,

Table 2
Correlation matrix of the parameters.

	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	LSF
SiO ₂	1				
Al ₂ O ₃	0.8001	1			
Fe ₂ O ₃	0.4217	0.5089	1		
CaO	-0.7787	-0.5969	-0.3464	1	
LSF	-0.8998	-0.7691	-0.4153	0.7148	1

respectively.

LM-ANN model

The LM-ANN was developed in MATLAB using its embedded neural network toolbox. The process began with the `nnstart` command, and the `nftool` (fitting app) was selected for input–output and curve fitting tasks. A two-layer feed-forward network with sigmoid hidden neurons and linear output neurons (`fitnet`) was employed. A total of Two thousand four hundred and sixty (2,460) input and output normalized datasets were used to construct the model. These datasets were randomly divided into training (70 %, 1,722 data points), validation (15 %, 369 data points), and testing (15 %, 369 data points). The LM-ANN architecture was structured as a 4-N-1 model, comprising one input layer, one hidden layer, and one output layer. The number of neurons in the hidden layer was determined iteratively from two (2) to eleven (11), as depicted in Table 3. For training the network, the Levenberg–Marquardt algorithm (`trainlm`) was utilized. A nonlinear tan-sigmoid (TANSIG) transfer function was applied to the hidden layer, while a linear (PURLIN) transfer function was used for the output layer. Subsequently, the simulation was executed.

PSO-ANN model

The PSO-ANN model was implemented within a MATLAB environment. The artificial neural network (ANN) architecture utilized a feed-forward structure and was trained using the particle swarm optimization (PSO) algorithm. Similar to the LM-ANN model, the datasets were normalized to a range of -1 to 1. A total of 2,460 input and output datasets were imported from the workspace to develop the model. To enhance the model's performance, the maximum iterations for the search agents were incrementally increased by 1,000, starting at 1,000. The model's performance, evaluated using the coefficient of correlation (R) and mean squared error (MSE), plateaued after 3,000 iterations. As a result, this value was adopted as the optimal number of iterations. A three-layer ANN structure was simulated, comprising an input layer, a hidden layer, and an output layer. The performance of each simulated ANN configuration was assessed using the coefficient of correlation (R) and the mean squared error (MSE) between the predicted outputs and the observed values. The optimal network architecture was determined through a trial-and-error approach, beginning with two neurons in the hidden layer and incrementally increasing by one neuron until the performance (R and MSE) stop improving.

GA-ANN model

The GA-ANN model was implemented in a MATLAB environment. Similar to the LM-ANN and PSO-ANN models, the datasets were normalized to a range of -1 to 1. A total of 2,460 input and output datasets were used to during the analysis. To develop the model, ANN architecture was defined, specifying the number of inputs, hidden neurons, and outputs. The ANN architecture employed a feed-forward network with five neurons in the hidden layer, based on insights gained from the LM-ANN and PSO-ANN models. Subsequently, the genetic algorithm (GA) optimization parameters, which regulate the GA process, were defined. These parameters included: Population size (PopSize): 150, number of generations (MaxGenerations): 50, Selection method: tournament selection, crossover probability (CrossPercent): 50 %, crossover ratio (Crossover): 15 %, crossover method: arithmetic crossover, mutation probability (MutatPercent): Calculated as $1 - \text{RecomPercent} - \text{CrossPercent}$, mutation method: Gaussian mutation, termination criteria: set at a fitness threshold. The fitness function was specified, and the initial population of potential solutions for the GA was created, where each individual represented a unique set of weights and biases for the ANN. The GA optimization process was then executed, iterating through selection, crossover, mutation, and evaluation steps for several generations until the population converged to an optimal solution representing optimized weights used to train the ANN.

Development of LSF predictive software application

To facilitate the practical application of the proposed models, the transformed mathematical equations were embedded into a software tool designed as a simple calculator for estimating the LSF based on the elemental composition of the cement raw mix. This application integrates three machine learning techniques LM-ANN, PSO-ANN, and GA-ANN within a unified platform, enabling users to select their preferred method and input relevant parameters to execute predictions. The graphical user interface (GUI) was

Table 3

LM-ANN simulations for LSF.

Networks	R			
	Training	Validation	Testing	Total
4-2-1	0.99554	0.98575	0.99760	0.99538
4-3-1	0.99625	0.99608	0.99612	0.99618
4-4-1	0.99525	0.99783	0.99695	0.99586
4-5-1	0.99545	0.99761	0.99708	0.99607
4-6-1	0.99528	0.99802	0.99600	0.99601
4-7-1	0.99481	0.99837	0.99702	0.99603
4-8-1	0.9956	0.99814	0.99335	0.9958
4-9-1	0.99702	0.98703	0.99528	0.9955
4-10-1	0.99476	0.99812	0.99382	0.9953
4-11-1	0.99742	0.98739	0.99832	0.9959

Bolded part indicate architecture with the highest training, validation, and testing R values.

developed using MATLAB's App Designer, following a structured three-stage process: GUI creation, GUI programming, and packaging for standalone deployment. These steps ensured that the software could function independently, offering users a seamless and efficient experience. A comprehensive description of the software development process is provided in Adamolekun et al. [20] and summarized in the next three subsections.

Creating the GUI

The process of creating the GUI involved selecting and assembling components from MATLAB's component library, such as numerical edit fields and buttons. The components used in this study included a panel, five numerical edit fields, and three buttons, all of which were selected from the library and added to the design view by a simple drag-and-drop action.

First, a panel was added to the design view, resized to occupy most of the blank space, and prepared to accommodate additional components. Next, five numerical edit fields were added to the panel. Four of these fields enabled users to input the parameters SiO_2 , Al_2O_3 , Fe_2O_3 , and CaO , while the fifth field displayed the estimated results generated by the software application. Additionally, three buttons were incorporated, each corresponding to one of the machine learning techniques (LM-ANN, PSO-ANN, and GA-ANN). These buttons were programmed to execute the calculations of the LSF based on the parameter values entered by the user.

To ensure clarity and ease of use, the GUI components were labeled appropriately: the input fields were labeled as SiO_2 , Al_2O_3 , Fe_2O_3 , and CaO , while the results field was labeled as LSF. The three buttons were labeled as CALCULATE LM-ANN, CALCULATE PSO-ANN, and CALCULATE GA-ANN. The final step in the process involved programming the GUI components to handle user interactions effectively.

Programming the GUI

Programming the Graphical User Interface (GUI) is crucial for developing user-friendly and interactive applications. It involves defining functions to process user inputs and application events, with a primary focus on using the push-button callback function. MATLAB simplifies GUI programming by automatically generating code for each component, which can be further customized. In this research, the functionalities of the CALCULATE LM-ANN, CALCULATE PSO-ANN, and CALCULATE GA-ANN buttons were implemented by integrating callback functions with the mathematical equations derived from the LM-ANN, PSO-ANN, and GA-ANN models, ensuring accurate execution of the intended calculations. These functions enabled the application to respond effectively to user interactions as they were seamlessly linked to the mathematical equations.

Packaging the application for standalone use

In this study, the GUI was packaged as a standalone application. This process involved two major stages: first, selecting the Designer and then the Share option, followed by choosing Standalone Desktop App from the share list. This triggered a compiler, which opened a new window (MATLAB Compiler).

In the second stage, a custom splash screen icon was selected and adjusted, followed by choosing the runtime environment. The option downloaded from Web was selected, and finally, the GUI was packaged as a standalone application. This approach allowed users to run the application independently, without requiring internet or server.

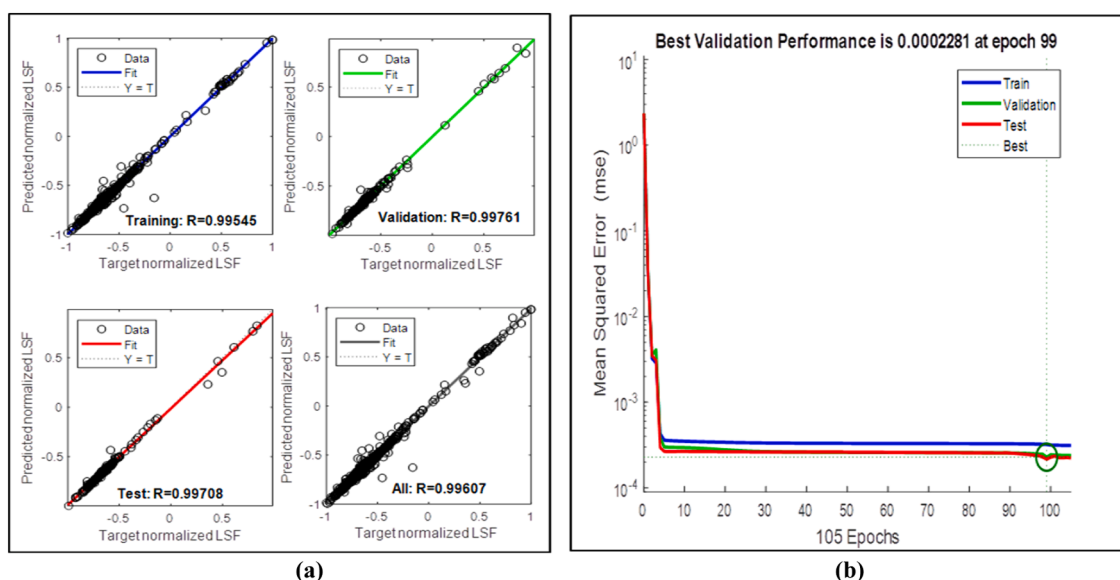


Fig. 3. Adopted LM-ANN model training, validation, and testing: (a) regression plots, (b) performance curve.

Results and discussions

This section presented the outcomes of this research; revealing novel insights into the research objectives, which yielded important conclusions regarding this research.

Adopted models

Proposed LS-ANN model

The training performance was assessed after each session by calculating the forecasting error between observed and predicted values. This evaluation encompassed the model's performance across the training, validation, and testing datasets, using both the coefficient of correlation (R) and the mean squared error (MSE) as metrics. If the model's performance was found to be unsatisfactory, the number of neurons in the hidden layer was increased and the network retrained. The optimal number of neurons was determined through a systematic trial-and-error approach, starting with two neurons in the hidden layer and incrementally increasing by one until no significant improvement in performance was observed for the total dataset. Beyond the 4-7-1 architecture (see Table 3), no network configuration achieved an R value equal to or greater than 0.99603 for the total dataset. Consequently, the training process was halted at the 4-11-1 architecture. The ANN configuration that delivered the highest overall performance indicated by the highest R value for the total dataset was adopted for accurate prediction of the LSF. In this study, the 4-5-1 architecture demonstrated superior performance (as highlighted in Table 3) and was selected as the final model.

The regression plot of the adopted LM-ANN model is presented in Fig. 3a. The R values for the training, validation, testing phases and the total datasets are approximately 0.99545, 0.99761, 0.99708, and 0.99607, respectively, indicating a strong positive correlation between the predicted and observed output values. Furthermore, the model's performance curve depicting mean squared error (MSE) against epochs as shown in Fig. 3b reveals that the training, validation, and testing datasets exhibit a consistent trend. This uniformity suggests that the model is not over fitted. Therefore, the adopted LM-ANN model can be considered robust and effective for accurate prediction.

Proposed PSO-ANN model

Ten different PSO-ANN models were simulated following the same procedure as the LM-ANN model, based on insights gained during its training specifically, the observation that network performance failed to improve beyond the 4-7-1 architecture, which led to the termination of the training at the 4-11-1 configuration. Like LM-ANN, the network architecture began with two neurons in the hidden layer, and one neuron was added incrementally until the 4-11-1 structure was reached. Each PSO-ANN model was evaluated using the R and MSE, as presented in Table 4. The optimal network was determined through the architecture that demonstrated the overall best performance characterized by the highest R and lowest MSE values. In this study, the 4-5-1 architecture (Fig. 4a) emerged as the top-performing configuration for predicting LSF, as illustrated in Table 4 (bolded entry).

The regression plot of the PSO-ANN model's predictions versus the measured LSF is shown in Fig. 4b. This visualization provides a clear assessment of the model's performance and accuracy. The plot indicates the model's success, with an R value of 0.99219, signifying a very strong positive correlation between the predicted and measured values. Additionally, the adopted PSO-ANN architecture's performance metrics, as illustrated in Table 4, report an MSE value of 0.00057743 for predicting the measured LSF. This exceptionally low error highlights the model's accuracy and reliability, reaffirming its effectiveness for practical applications.

Proposed GA-ANN model

The optimal network architecture for the GA-ANN model was identified through a systematic trial-and-error process, beginning with two neurons in the hidden layer and incrementally increasing by one neuron up to eleven consistent with the approach used for the LM-ANN and PSO-ANN models. Each network configuration was evaluated based on its performance metrics, specifically the R and MSE, as presented in Table 5. The architecture that exhibited the best combination of the highest R value and the lowest MSE was selected as the optimal model. As shown in Table 5, the GA-ANN model with a 4-3-1 architecture achieved the highest R value (0.9691) and the lowest MSE (0.0025), making it the most effective configuration for predicting LSF in this study.

Table 4
ANN simulations for PSO-ANN.

Networks	R	MSE
4-2-1	0.97059	0.0022
4-3-1	0.93548	0.0046
4-4-1	0.9883	0.000863
4-5-1	0.99219	0.000577
4-6-1	0.97084	0.0021
4-7-1	0.99037	0.000712
4-8-1	0.97714	0.0017
4-9-1	0.99046	0.000709
4-10-1	0.98056	0.0014
4-11-1	0.99042	0.000707

Bolded part indicate network with the highest R and lowest MSE values.

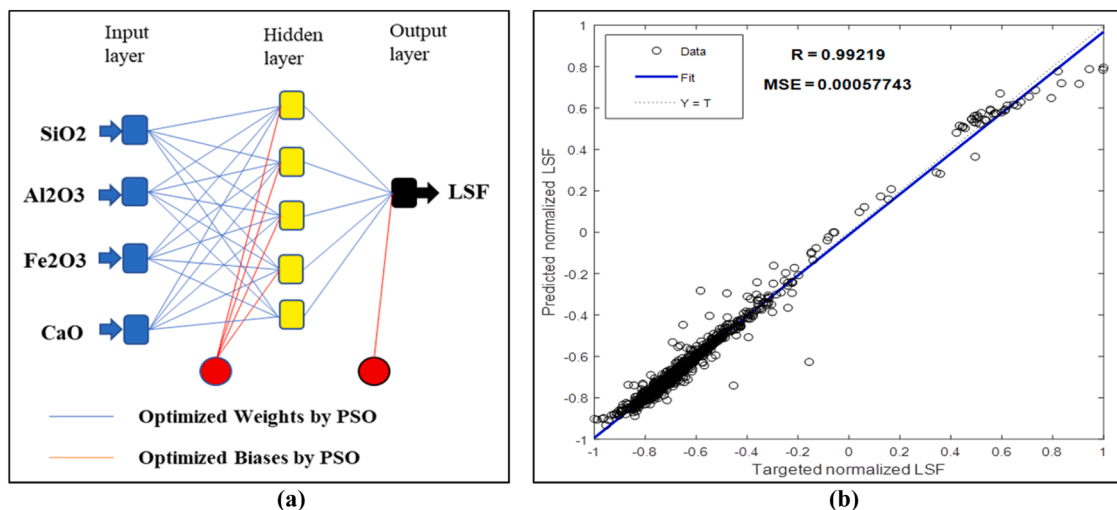


Fig. 4. Adopted PSO-ANN modelling framework: (a) Adopted PSO-ANN architecture, (b) Regression plot.

Table 5
ANN Simulations for GA-ANN.

Networks	R	MSE
4-2-1	0.9650	0.0028
4-3-1	0.9691	0.0025
4-4-1	0.9553	0.0033
4-5-1	0.9420	0.0037
4-6-1	0.9572	0.0037
4-7-1	0.9582	0.0036
4-8-1	0.9690	0.0027
4-9-1	0.9626	0.0030
4-10-1	0.9539	0.0036
4-11-1	0.9640	0.0026

Bolded part indicate network with the highest R and lowest MSE values.

Fig. 5a highlights the adopted GA-ANN regression plot, showcasing a strong positive correlation between the predicted and measured LSF values. This demonstrates the model's robust predictive accuracy and reliability, confirming the suitability of the proposed GA-ANN model for practical applications.

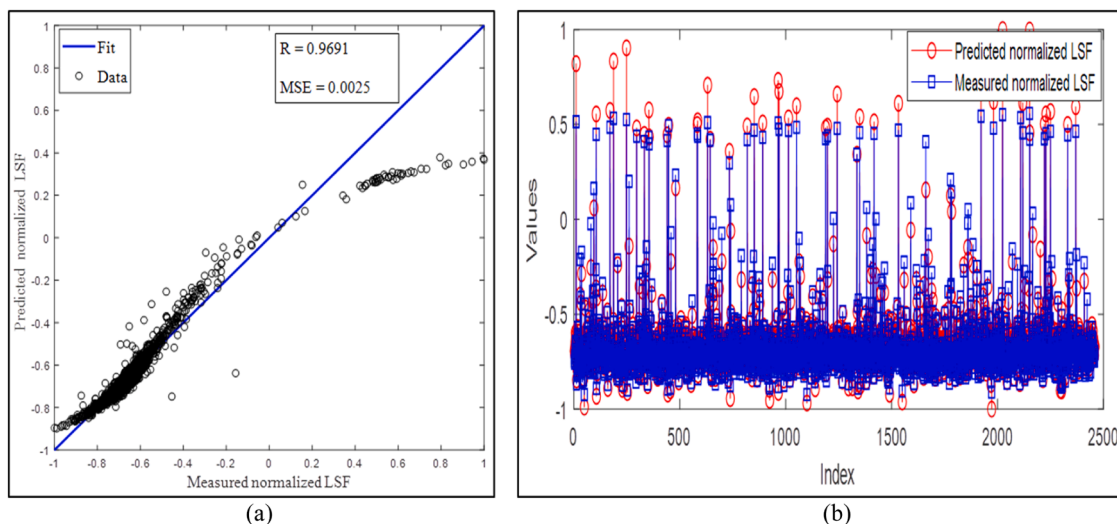


Fig. 5. Adopted GA-ANN modelling framework: (a) Regression plot, (b) Index plot.

The performance of the adopted GA-ANN model was further evaluated using an index plot of measured versus predicted values, as presented in Fig. 5b. This graphical representation serves as a tool for assessing how effectively the model predicts a given outcome. It involves plotting the measured (observed) values and the predicted values against their respective indices (typically the sequence in which the data points appear). This approach provides insights into the accuracy of predictions, patterns and trends in the data, and any deviations between the measured and predicted values. The plot (Fig. 5b) reveals that the distribution patterns of the measured values and the model predictions exhibit a significant overlap, with only minor deviations. This indicates a strong relationship between the measured and predicted LSF values, demonstrating a high level of accuracy and reliability in the model's performance. Consequently, the GA-ANN model is considered successful and dependable for practical applications.

To enhance usability and simplicity, the proposed models were transformed from a "black box" representation into a "white box" form, meaning they were converted into mathematical models. This transformation is discussed in detail in the following subsection.

Transformation of proposed models to simple mathematical equations

The major drawback of most of the existing ANN models for predicting LSF is that they were not converted into mathematical equations. As a result, many users find them difficult to apply. In this study, the three developed hybrid models were converted into mathematical equations to ensure ease of application. To achieve this, they were transformed through the weights and biases based on the ANN general equation presented in Eq. (2). The mathematical formulas obtained from the LM-ANN, PSO-ANN and GA-ANN models are presented in Eqs. (3)–(8), respectively.

$$y = f_{\text{purlin/sig}} \left\{ b_0 + \sum_{k=1}^n \left[f_{\text{sig}} \left(b_{bk} + \sum_{i=1}^m w_{ik} x_i \right) \times w_k \right] \right\} \quad (2)$$

where b_0 is the bias in the output layer; w_k is the weight of the connection between the k^{th} of the hidden layer and the single output neuron; b_{bk} is the bias in the k^{th} neuron of the hidden layer; m is the number of neurons in the input layer; n is the number of neurons in the hidden layer; w_{ik} is the weight of the connection between the i^{th} input parameter and the hidden layer; x_i is the input variable i ; y is the output variable; f_{purline} and f_{sig} are the transfer functions, which are linear and nonlinear transfer functions, respectively.

For the case of LM-ANN and PSO-ANN, f_{purline} is employed as the output transfer function, whereas f_{sig} is utilized for GA-ANN. For easy identification, LSF predicted by LM-ANN, PSO-ANN, and GA-ANN are represented as $LSF_{\text{LM-ANN}}$, $LSF_{\text{PSO-ANN}}$, $LSF_{\text{GA-ANN}}$, respectively.

LM-ANN model derived mathematical equation is shown in Eq. (3).

$$LSF_{\text{LM-ANN}} = 191.47 (1.655208 - 2.99706 x_1 + 0.034469 x_2 - 1.52641 x_3 + 0.115768 x_4 - 1.9159 x_5) + 247.1 \quad (3)$$

where

$$\begin{aligned} x_1 &= \tanh (4.02054 + 0.33251 \text{ SiO}_2 + 0.09597 \text{ Al}_2\text{O}_3 - 0.16211 \text{ Fe}_2\text{O}_3 - 0.08236 \text{ CaO}) \\ x_2 &= \tanh (-10.39961 - 0.12411 \text{ SiO}_2 + 0.13333 \text{ Al}_2\text{O}_3 + 0.31766 \text{ Fe}_2\text{O}_3 + 0.25025 \text{ CaO}) \\ x_3 &= \tanh (-0.46515 + 0.09373 \text{ SiO}_2 + 0.00096 \text{ Al}_2\text{O}_3 + 0.03467 \text{ Fe}_2\text{O}_3 - 0.0065 \text{ CaO}) \\ x_4 &= \tanh (-0.31404 - 0.27675 \text{ SiO}_2 - 0.39293 \text{ Al}_2\text{O}_3 - 0.3257 \text{ Fe}_2\text{O}_3 + 0.06963 \text{ CaO}) \\ x_5 &= \tanh (4.09207 - 0.24574 \text{ SiO}_2 + 0.16703 \text{ Al}_2\text{O}_3 - 0.05055 \text{ Fe}_2\text{O}_3 - 0.09845 \text{ CaO}) \end{aligned}$$

For easy application, Eq. (3) is merged as shown in Eq. (4).

$$LSF_{\text{LM-ANN}} = 191.47 (1.655208 - 2.99706 \tanh (4.02054 + 0.33251 \text{ SiO}_2 + 0.09597 \text{ Al}_2\text{O}_3 - 0.16211 \text{ Fe}_2\text{O}_3 - 0.08236 \text{ CaO}) + 0.034469 \tanh (-10.39961 - 0.12411 \text{ SiO}_2 + 0.13333 \text{ Al}_2\text{O}_3 + 0.31766 \text{ Fe}_2\text{O}_3 + 0.25025 \text{ CaO}) - 1.52641 \tanh (-0.46515 + 0.09373 \text{ SiO}_2 + 0.00096 \text{ Al}_2\text{O}_3 + 0.03467 \text{ Fe}_2\text{O}_3 - 0.0065 \text{ CaO}) + 0.115768 \tanh (-0.31404 - 0.27675 \text{ SiO}_2 - 0.39293 \text{ Al}_2\text{O}_3 - 0.3257 \text{ Fe}_2\text{O}_3 + 0.06963 \text{ CaO}) - 1.9159 \tanh (4.09207 - 0.24574 \text{ SiO}_2 + 0.16703 \text{ Al}_2\text{O}_3 - 0.05054 \text{ Fe}_2\text{O}_3 - 0.09845 \text{ CaO})) + 247.1 \quad (4)$$

For the case of PSO-ANN model, the transformed mathematical equation is presented in Eq. (5),

$$LSF_{\text{PSO-ANN}} = 191.47 (0.094867 - 0.15113 y_1 + 0.173096 y_2 + 0.54802 y_3 + 0.053117 y_4 - 0.50802 y_5) + 247.1 \quad (5)$$

where

$$\begin{aligned} y_1 &= \tanh (0.094867 + 1.189425 \text{ SiO}_2 - 1.08145 \text{ Al}_2\text{O}_3 + 1.5 \text{ Fe}_2\text{O}_3 - 0.44315 \text{ CaO}) \\ y_2 &= \tanh (1.5 + 0.74764 \text{ SiO}_2 - 1.5 \text{ Al}_2\text{O}_3 - 0.17172 \text{ Fe}_2\text{O}_3 - 1.5 \text{ CaO}) \\ y_3 &= \tanh (-1.5 - 1.37718 \text{ SiO}_2 + 0.427046 \text{ Al}_2\text{O}_3 + 0.319555 \text{ Fe}_2\text{O}_3 + 1.5 \text{ CaO}) \\ y_4 &= \tanh (-1.49952 + 1.002422 \text{ SiO}_2 - 1.5 \text{ Al}_2\text{O}_3 + 0.212273 \text{ Fe}_2\text{O}_3 + 0.212277 \text{ CaO}) \\ y_5 &= \tanh (1.024655 + 1.281054 \text{ SiO}_2 + 0.976167 \text{ Al}_2\text{O}_3 - 0.68964 \text{ Fe}_2\text{O}_3) + 0.359931 \text{ CaO}) \end{aligned}$$

For easy usage, Eq. (5) is merged as shown in Eq. (6).

$$LSF_{\text{PSO-ANN}} = 191.47 (0.094867 - 0.15113 \tanh (0.094867 + 1.189425 \text{ SiO}_2 - 1.08145 \text{ Al}_2\text{O}_3 + 1.5 \text{ Fe}_2\text{O}_3 - 0.44315 \text{ CaO})) + 0.173096 \tanh (1.5 + 0.74764 \text{ SiO}_2 - 1.5 \text{ Al}_2\text{O}_3 - 0.17172 \text{ Fe}_2\text{O}_3 - 1.5 \text{ CaO})) + 0.54802 \tanh (-1.5 - 1.37718 \text{ SiO}_2 + 0.427046 \text{ Al}_2\text{O}_3 + 0.319555 \text{ Fe}_2\text{O}_3 + 1.5 \text{ CaO})) + 0.053117 \tanh (-1.49952 + 1.002422 \text{ SiO}_2 - 1.5 \text{ Al}_2\text{O}_3 + 0.212273 \text{ Fe}_2\text{O}_3 + 0.212277 \text{ CaO}) - 0.50802 \tanh (1.024655 + 1.281054 \text{ SiO}_2 + 0.976167 \text{ Al}_2\text{O}_3 - 0.68964 \text{ Fe}_2\text{O}_3) + 0.359931 \text{ CaO})) + 247.1 \quad (6)$$

For the case of PSO-ANN model, the transformed mathematical equation is presented in Eq. (7),

$$LSF_{GA-ANN} = 191.47 \tanh (-0.25437 + 0.275608z_1 + 0.490303z_2 - (0.5975z_3) + 247.1 \quad (7)$$

where

$$z_1 = \tanh (-0.01917 - 0.45246 \text{SiO}_2 - 0.323 \text{Al}_2\text{O}_3 - 0.18133 \text{Fe}_2\text{O}_3 - 0.04216 \text{CaO})$$

$$z_2 = \tanh (-0.51809 - 0.62699 \text{SiO}_2 - 0.2911 \text{Al}_2\text{O}_3 + 0.256197 \text{Fe}_2\text{O}_3 + 0.372276 \text{CaO})$$

$$z_3 = \tanh (0.7587481 + 0.476442 \text{SiO}_2 + 0.055546 \text{Al}_2\text{O}_3 + 0.510062 \text{Fe}_2\text{O}_3 - 0.4044 \text{CaO})$$

For streamline, Eq. (7) is merged as shown in Eq. (8).

$$LSF_{GA-ANN} = 191.47 \tanh (-0.25437 + 0.275608 \tanh (-0.01917 - 0.45246 \text{SiO}_2 - 0.323 \text{Al}_2\text{O}_3 - 0.18133 \text{Fe}_2\text{O}_3 - 0.04216 \text{CaO}) + 0.490303 \tanh (-0.51809 - 0.62699 \text{SiO}_2 - 0.2911 \text{Al}_2\text{O}_3 + 0.256197 \text{Fe}_2\text{O}_3 + 0.372276 \text{CaO}) - 0.5975 \tanh (0.7587481 + 0.476442 \text{SiO}_2 + 0.055546 \text{Al}_2\text{O}_3 + 0.510062 \text{Fe}_2\text{O}_3 - 0.4044 \text{CaO}) + 247.1 \quad (8)$$

ANNs are known for generating complex mathematical formulations due to their layered architecture, which captures intricate relationships between input and output variables. Each layer consists of interconnected neurons, where each neuron performs a mathematical operation. These operations collectively form a nested composite function, resulting in highly nonlinear and sophisticated equations. This structural complexity underpins the superior predictive performance of ANN models, as the inclusion of multiple terms enables the capture of nuanced patterns that simpler models may overlook. To streamline the representation of the model into a single-step form, Eqs. (3), (5), and (7) were consolidated into Eqs. (4), (6), and (8), respectively. This merging facilitates seamless integration into elemental composition analyzers such as CBA and XRF in cement industry. Furthermore, the simplified equations enable rapid LSF estimation using common spreadsheet tools like Microsoft Excel, enhancing accessibility for cement industry practitioners. However, prior to practical deployment, it is essential to validate these equations to ensure their reliability. The validation process is comprehensively detailed in the futher subsection.

Validation of the proposed models derived mathematical equations

The performance of the mathematical equations presented in Eqs (3) and (8) must be benchmarked against the original ANN model to validate their predictive accuracy. To achieve this, the predictions generated directly by the LM-ANN, PSO-ANN, and GA-ANN models were compared with those obtained from Eqs. (3) to (8) using verification dataset comprising 100 data points, as illustrated in Fig. 6.

It was shown in Fig. 6 that the coefficient of determination (R^2) from plotting the computed LM-ANN, PSO-ANN and GA-ANN output against the direct models output was 1, indicating that the mathematical equations explained 100 % of the variation in the direct LM-ANN, PSO-ANN and GA-ANN models predictions. Therefore, the equations are successful and can be directly applied for practical purposes.

To further enhance usability, simplicity, and user-friendliness, the mathematical equations were integrated into a software application designed as a simple calculator, as discussed in the subsequent subsection.

Developed software application

Computational errors may arise when using mathematical equations, especially among end users with limited mathematical skills. Moreover, the time required to solve such equations often depends on the user's level of proficiency. To overcome these limitations,

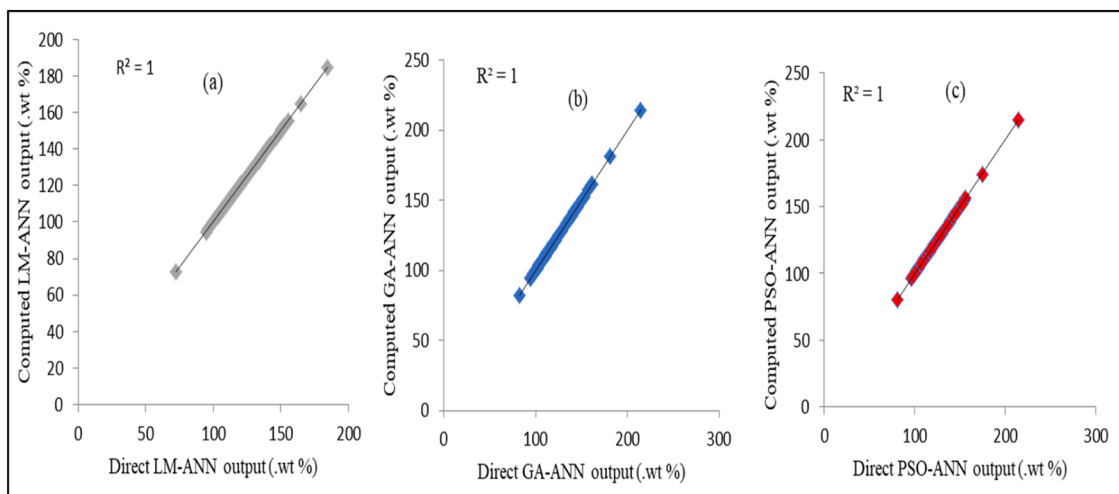


Fig. 6. Comparison of the mathematical equations and direct models simulated outputs of: (a) LM-ANN, (b) GA-ANN and (c) PSO-ANN.

this study developed a software application that embeds the ANN-transformed mathematical equations, functioning as a simple calculator for quick and accurate estimation of the LSF. The user interface of the application is presented in Fig. 7.

The software is designed as a three-in-one application integrating LM-ANN, PSO-ANN, and GA-ANN models, allowing users to select their preferred option for LSF estimation. User interaction with the application will be by entering the oxide composition values (SiO_2 , Al_2O_3 , Fe_2O_3 , and CaO) into the designated input fields and clicking any of the CALCULATE LM-ANN, CALCULATE PSO-ANN, or CALCULATE GA-ANN buttons, depending on their choice. Within approximately one second, the application processes the input parameters and displays the estimated LSF value in the output field. This functionality enables users to easily compare the predictions from the three models and make informed decisions based on the comparison. The integrated software model will provide an efficient and reliable means of assessing and maintaining the performance and integrity of CBA and XRF systems in cement production. To ensure the models generalize well to new data and produce consistent and trustworthy predictions, they were validated as described in the next subsection.

Models validation

To ensure that a model's performance is robust and generalizable, it is essential to validate the model. One commonly employed validation technique is cross-validation, which has been extensively used by researchers in the past. Cross-validation is a method in data analysis and machine learning that evaluates a model's performance by splitting the dataset into multiple parts (or "folds"). The model is then trained and tested on these parts to confirm its ability to generalize to unseen data. Typically, the dataset is divided into a specified number of folds, often 5 or 10, in a method known as k-fold cross-validation. The model is trained on k-1 folds and tested on the remaining fold. This process is repeated until each fold has been used as a test set exactly once. Performance metrics such as R^2 , MAPE, and MAE are averaged across all folds to provide an overall evaluation. This approach helps to identify over-fitting or under-fitting while ensuring consistent performance across different subsets of the data. In this research, the models were validated using a 5-fold cross-validation procedure with random sampling, employing performance metrics such as R^2 , RMSE, and MAE as illustrated in Tables 6–8.

For the LM-ANN model (Table 6), the average R^2 , MAPE, and MAE values were 0.9815, 0.9342, and 1.1625, respectively. All the R^2 values are extremely high, indicating excellent model fit and predictive power. Also, the average R^2 value suggests a very high level of accuracy. Furthermore, the standard deviation of the R^2 values is very low (0.015), indicating minimal variation in model performance across the folds. For the case of MAPE and MAE, the results show that the values are relatively low, indicating good model performance. Also, the standard deviation of the MAPE and MAE values is 0.43 and 0.45, respectively, indicating very slight variation in model performance across the fold; providing accurate predictions with minor error. This consistency in the performance validates the

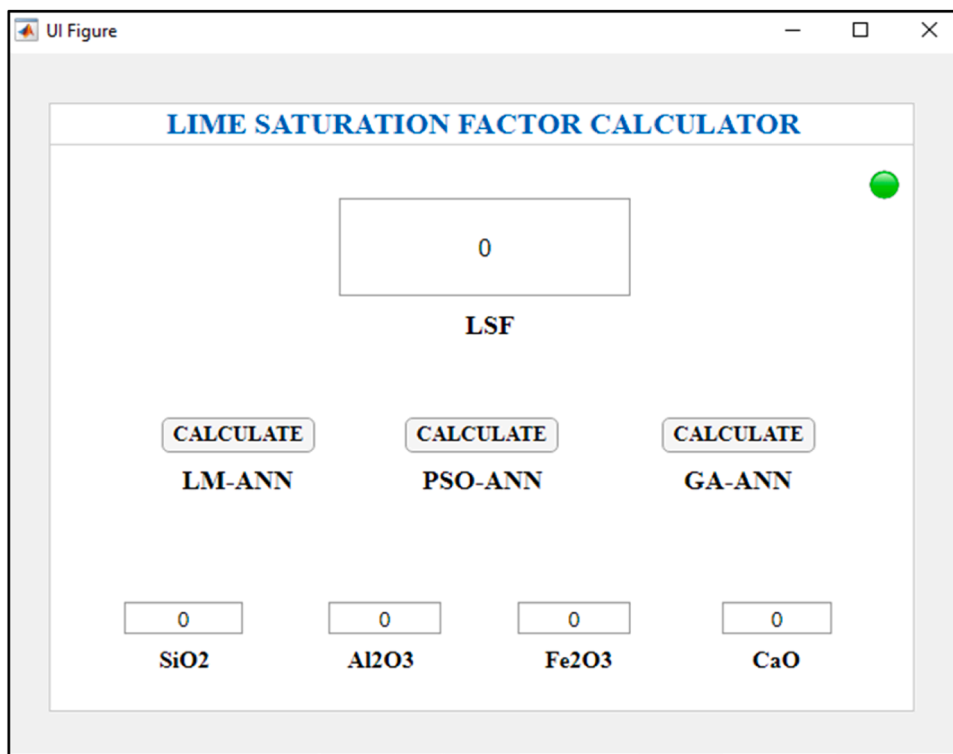


Fig. 7. User interface of the ANN-based standalone GUI application.

Table 6
5-Fold cross-validation performance for LM-ANN.

Statistical Analysis	Folds					Average
	1	2	3	4	5	
R ²	0.99	0.952	0.9778	0.9904	0.9971	0.9815
MAPE	0.7231	1.3817	1.4011	0.6545	0.511	0.9342
MAE	0.8772	1.7192	1.7883	0.7324	0.6952	1.1625

Table 7
5-Fold cross-validation performance for PSO-ANN.

Statistical Analysis	Folds					Average
	1	2	3	4	5	
R ²	0.9596	0.94	0.9494	0.9837	0.9867	0.9639
MAPE	1.01	2.3182	2.4069	1.754	1.805	1.8588
MAE	1.2724	2.8198	2.791	1.8326	1.7638	2.0959

Table 8
5-Fold cross-validation performance for GA-ANN.

Statistical Analysis	Folds					Average
	1	2	3	4	5	
R ²	0.911	0.9191	0.943	0.9733	0.9658	0.9424
MAPE	3.2323	3.2338	4.4088	1.9584	4.068	3.3803
MAE	3.9029	4.0778	5.6981	2.4447	7.7527	4.7752

use of 5-fold cross-validation and confirms that the model is reliable and suitable for practical applications.

For the case of PSO-ANN, the performance of the model is comparable to that of the LM-ANN model (Table 7). The average R² value across the 5 folds is 0.9639, with the range of values typically within 3 % of the average. This consistency indicates that the model's performance is stable across different folds of cross-validation. Additionally, the average error metrics, MAPE and MAE, are 1.8588 and 2.0959, respectively. These small error values further emphasize the model's reliability. The deviation of the error in each fold from the average is minimal; ranging between 0.05 and 0.55 for MAPE and between 0.26 and 0.72 for MAE. This low deviation serves as additional evidence of the model's stability, suggesting that it neither over fits nor under fits the data. Therefore, it is validated.

The R² and prediction errors of the adopted GA-ANN model for the prediction of LSF across the 5-folds are presented in Table 8. The model achieved high R² values across the folds, with the lowest being 0.911, indicating excellent model fit. Furthermore, the average R² is approximately 0.9424 which suggests a high level of model accuracy. Also, the standard deviation of the R² values across the 5 folds is relatively low (approximately 0.025), indicating consistent model performance across the 5 folds. The error metrics (MAPE and MAE) across the 5 folds are small and very close to the average values. The average MAPE value is 3.38, which suggests a high level of accuracy. The standard deviation of MAPE is 1.04, indicating some variation in model performance across the folds. For the case of MAE, the standard deviation is 1.73, indicating a moderate level of variation in model performance across the folds. This minimal variation demonstrates the reliability of the 5-fold cross-validation results. In summary, based on the high R², MAPE and MAE low standard deviation, the 5-fold validation is successful in confirming the model's accuracy and reliability.

In summary, the three developed models (LM-ANN, PSO-ANN, and GA-ANN) effectively captured the nonlinear relationship between the input variables and the target output, demonstrating high predictive accuracy. Their performance is robust and generalizable. Thus, they are all validated and can be used for practical purpose.

To identify the most effective model for predicting LSF, it is necessary to compare the various models both with one another and against Bogue's model. A detailed discussion of this comparison is provided in the further subsection.

Models comparison and selection

In prior studies, several mathematical models have been utilized for the estimation of lime saturation factor (LSF). Among these, the most prominent is the Bogue's mathematical model, presented in Eq. (9), which is widely adopted in the cement industry. This research seeks to develop a model with higher accuracy than the Bogue's [8] model, which serves as the benchmark for comparison.

$$\text{LSF (\%)} = \frac{\text{CaO}}{2.8 \text{ SiO}_2 + 1.18 \text{ Al}_2\text{O}_3 + 0.65 \text{ Fe}_2\text{O}_3} \times 100 \quad (9)$$

To evaluate the proposed models, a new trial dataset comprising one hundred (100) data points was employed. This dataset was used to assess the models' efficiency against the Bogue's model and their ability to predict beyond the training data using a set of standard prediction performance indices, including R², RMSE, MAPE, MAE and RSE as presented in Eqs. (10) to (15).

$$R^2 = 1 - \frac{\sum_{i=1}^n (E - P)^2}{\sum_{i=1}^n (E - \bar{Y})^2} \quad (10)$$

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n E \quad (11)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (E - P)^2} \quad (12)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{(E - P)}{E} \times 100 \quad (13)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |E - P| \quad (14)$$

$$RSE = \sqrt{\frac{1}{n - m - 1} \sum_{i=1}^n (E - P)^2} \quad (15)$$

where, n is the number of sample data points used for model development, m is the number of predictors (independent variables), and E and P represent the measured and predicted values of engineering properties, respectively.

R-squared analysis

The models were employed to predict LSF values using 100 input parameters from the trial dataset. The predicted LSF values were then compared to the measured ones by plotting them against each other. The results, evaluated based on R^2 , are illustrated in Fig. 8. A higher R^2 value signifies better model performance.

According to the results (Fig. 8), the proposed LM-ANN model achieved the highest R^2 value ($R^2 = 0.9885$) compared to the Bogue's model ($R^2 = 0.9867$), PSO-ANN ($R^2 = 0.9813$), and GA-ANN ($R^2 = 0.9603$). This implies that the LM-ANN model explains 98.85 % of the variability in the measured LSF values, surpassing the Bogue's model, which accounts for 98.67 %. Likewise, PSO-ANN captures 98.13 % of the variability, while GA-ANN explains 96.03 %. Among all the models assessed in this study, only the LM-ANN model outperformed the Bogue's model, highlighting its superior estimation accuracy over the benchmark model. To select the best model convincingly, the models were further compared using error analysis. The detailed discussion is presented in the next subsection.

Error analysis

The prediction errors of the proposed LM-ANN, PSO-ANN, GA-ANN models, as well as the Bogue's model, were analyzed using RMSE, MAPE, MAE, and RSE, as presented in Table 9 and Figs. 9a and b. A lower error magnitude in these metrics indicates better

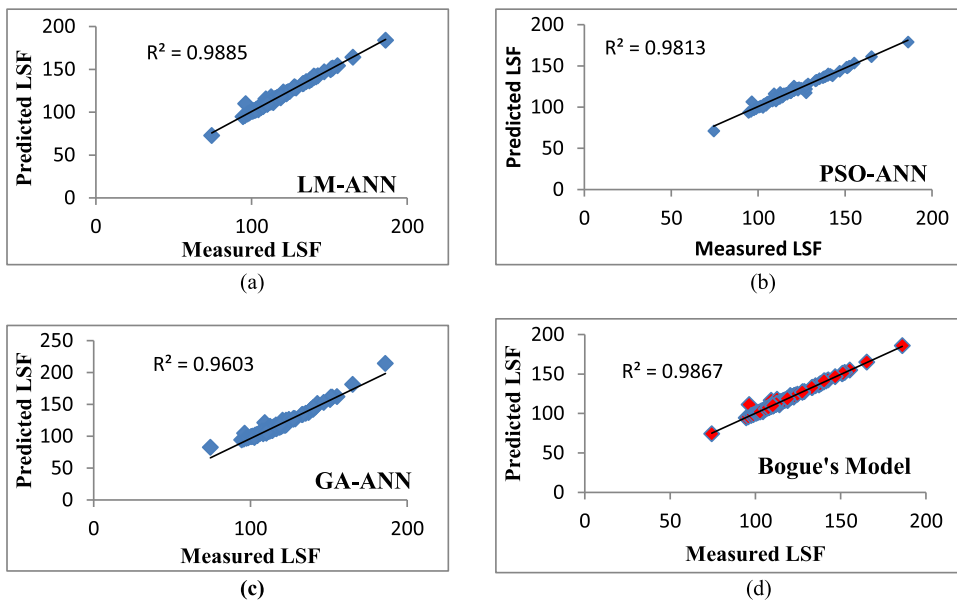
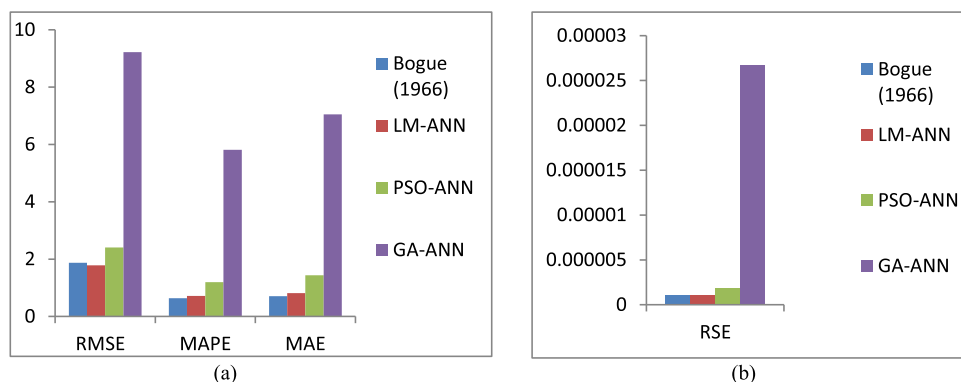


Fig. 8. Regression plots for: (a) LM-ANN; (b) PSO-ANN; (c) GA-ANN; and (d) Bogue's Model.

Table 9Error analysis for the MDD, OMC and CBR_U adopted models.

Proposed Model	Statistical analysis			
	RMSE	MAPE	MAE	RSE
Bogue [8]	1.8678	0.6309	0.7031	1.1E-06
LM-ANN	1.7828	0.7186	0.8115	9.99E-07
PSO-ANN	2.4043	1.1953	1.4364	1.82E-06
GA-ANN	4.9275	2.7326	3.3530	7.63E-06

**Fig. 9.** Models performance comparison using Verification dataset for: (a) RMSE, MAPE, MAE; and (b) RSE.

model performance.

Table 9 and Fig. 9a reveal that the LM-ANN model had the lowest RMSE value (1.7828), followed by the Bogue's model (1.8678), PSO-ANN (2.4043), and lastly, GA-ANN (4.9275). For MAPE, the Bogue's model achieved the lowest value (0.6309), followed by LM-ANN (0.7186), PSO-ANN (1.1953), and GA-ANN (2.7326). Similarly, in the case of MAE, the Bogue's model displayed the best performance with the lowest value (0.7031), followed by LM-ANN (0.8115), PSO-ANN (1.4364), and GA-ANN (3.3530). Regarding RSE (Table 9, Fig. 9b), the LM-ANN model demonstrated the lowest value (9.99E-07), with the Bogue's model in second place (1.1E-06), followed by PSO-ANN (1.82E-06), and GA-ANN occupying the last spot (7.63E-06). Based on the error analysis, the LM-ANN and Bogue's model displayed comparable performances. Each model achieved the lowest error in two metrics: LM-ANN excelled in RMSE and RSE, while the Bogue's model performed best in MAPE and MAE.

In summary, the LM-ANN model outperformed other proposed models in all the metrics, and the Bogue's model in three metrics, R², RMSE, and RSE; while the Bogue's model excelled in two metrics, MAPE and MAE. It can be concluded that the LM-ANN model demonstrates superior LSF prediction compared to the Bogue's model. Therefore, the LM-ANN model is strongly recommended for accurate prediction of LSF, having exhibited better performance overall. The mathematical formula (Eq. 4) of the LM-ANN model can be integrated into elemental composition analyzers, such as CBA and XRF. This integration will enable real-time, data-driven adjustments during cement production, empowering plant operators to:

- optimize phase composition control, reducing undesired compounds like free lime and excess alite;
- energy efficiency in kilns, leading to significant reductions in fuel consumption and thermal energy waste;
- reduce cost of cement production;
- reduced kiln refractory wear, leading to longer refractory life and lower maintenance costs;
- enhance raw material utilization, lowering variability and improving blend consistency; and
- support sustainable manufacturing, by reducing carbon emissions and material loss.

Furthermore, the developed software application will enable regular monitoring of the reliability of elemental composition analyzers (CBA or XRF) in standard cement production processes. Reliability checks help identifies sensor drift, calibration errors, or hardware malfunction before they cause significant production losses or off-spec cement. Also, in a small-scale cement plants or laboratories without access to CBA or XRF, which rely on wet chemical analysis and traditional oxide determination techniques, can use the software application for quick estimation of LSF. Ultimately, this research advances the precision of LSF estimation, directly contributing to the production of higher-quality cement. The will helps fortify infrastructure performance and significantly reduce the risk of premature failure and loss of life.

Sensitivity analysis

Sensitivity analysis is a technique used to evaluate how changes in input variables influence variations in the output variable [20].

Several methods can be employed for sensitivity analysis, including the one-at-a-time approach, scenario analysis, Fourier amplitude sensitivity test (FAST), and the Cosine amplitude method (CAM). Among these, the CAM has gained significant popularity among researchers [18,22,23] due to its reliability. The CAM has been extensively utilized in the field of geotechnical engineering [18,20,22–24], where it has consistently demonstrated excellent performance. For this study, CAM was adopted to identify the significance of the input parameters (SiO_2 , Al_2O_3 , Fe_2O_3 , and CaO) on the LSF estimated by the LM-ANN-based software application, as presented in Eq. (16).

$$R_{ij} = \frac{\sum_{k=1}^n (I \times P)}{\sqrt{\sum_{k=1}^n I^2 \sum_{i=1}^n P^2}} \quad (16)$$

where R_{ij} represents the strength of the input parameter, I represents the model input, P is the predicted output, and n is the number of data points. The result of the sensitivity is presented in Fig. 10.

The analysis of the input parameters' significance, illustrated in Fig. 10 based on the CAM approach in Eq. (16), revealed that CaO had the highest influence on LSF, followed by SiO_2 , Fe_2O_3 , and Al_2O_3 . In summary, the results of the sensitivity analysis indicated that CaO was the most influential input variable on LSF. This finding aligns with the results of the correlation matrix, which also identified CaO as having the highest correlation with LSF.

Conclusions

This research highlights the pivotal role of LSF in cement quality assurance and infrastructure resilience. While the Bogue [8] formula has historically served as the benchmark for LSF estimation, its limitations hinder adaptability to modern production dynamics. In contrast, the machine learning-based models developed in this study: LM-ANN, PSO-ANN, and GA-ANN, demonstrated enhanced predictive accuracy and robustness using a comprehensive dataset from Dangote Cement Plants across Nigeria.

Among these models, LM-ANN consistently outperformed the others across key performance metrics, including R^2 , RMSE, and RSE, while maintaining competitive results in MAPE and MAE. The LM-ANN model thus represents a significant advancement in LSF prediction, offering a scalable and intelligent alternative to traditional methods. Integrating the developed mathematical model into CBA and XRF systems will facilitate accurate real-time estimation and control of LSF, enabling immediate adjustments in raw mix proportions to maintain optimal clinker composition. This integration will minimize process variability, enhance cement quality consistency, reduce energy consumption, and promote data-driven, efficient production control.

The software application will facilitate rapid estimation of LSF and accurate monitoring of CBA and XRF system performance and integrity within the cement industry. Sensitivity analysis identified CaO as the most influential input variable, reaffirming the model's cement chemistry and process optimization.

However, the study is limited by its reliance on regional data and the exploration of only three optimization algorithms (LM, PSO, and GA). Future research should consider hybrid or ensemble models that combine ANN with other optimization techniques and expand the dataset to encompass diverse geological sources and production conditions.

Author's contributions

Lateef Bankole Adamolekun wrote the first draft of the manuscript and performed the material preparation, data collection and analysis, and implemented the models development and validation, and software application development, provided resources, supervision and project administration.

Availability of data

The dataset generated during the current study are available at <https://doi.org/10.6084/m9.figshare.28791806>.

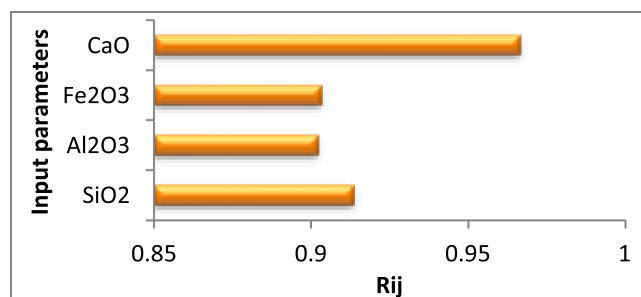


Fig. 10. Strength of the input parameters on the predicted LSF.

Funding

The research received no external funding.

Competing interest

The author has no relevant financial and non-financial interests to disclose.

Ethical approval

The authors state that the research was conducted according to ethical standards.

Declarations of generative AI-assisted technologies in the writing process

During the preparation of this work, the authors used [American Journal Experts] in order to improve readability and language. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The author declares that they have no conflicts of interest.

References

- [1] N. Sanusi, Effect of free lime and lime saturated factor on grindability of cement clinker, *Int. J. Eng. Res. Rev.* 7 (1) (2020) 61–66.
- [2] B.O. Taiwo, M.N. Shahahi, A. Omosebi, O.B. Samson, A.A. Akinlabi, Development of mathematically motivated artificial intelligence models for the prediction of carbonate rock lime saturation factor for cement production, *Eng. Appl. Artif. Intell.* 127 (Part A) (2024), <https://doi.org/10.1016/j.engappai.2023.107444>.
- [3] M. Irfan, D. Wiwik, Subandrio, A.A. Masagus, K. Riskaviana, F. Irsan, Variogram modeling of lime saturation factor on limestone quarry, *Int. J. Sci. Technol. Res.* 9 (3) (2020) 1679–1684.
- [4] S. Boughanmi, I. Labidi, H. Tiss, A. Megriche, The effect of marl and clay composition on the Portland cement quality, *J. Tunis. Chem. Soc.* 18 (2016) 43–51.
- [5] N. Sanusi, A.B. Nafiu, A.M. Umar, Effect and control of free lime and lime saturated factor of raw materials for clinker production, *Int. J. Adv. Acad. Res.* 8 (7) (2022) 116–124.
- [6] R. Kumar, V. Kumar, Prediction of lime saturated factor in cement raw mix using multiple linear regression, *J. Cem. Concr. Res.* 6 (1) (2018) 1–9.
- [7] A.E. Tiryaki, R. Kozan, N.G. Adar, Mathematical modeling of a cement raw-material blending process using a neural network, *Mater. Technol.* 50 (4) (2015) 485–490.
- [8] R.H. Bogue, The chemistry of Portland cement, 2nd ed., Reinhold Publishing Corporation, New York, 1966.
- [9] A.I. Lawal, An artificial neural network-based mathematical model for the prediction of blast-induced ground vibration in granite quarries in Ibadan, Oyo State Nigeria, *Sci. Afr.* 8 (2020) e00413.
- [10] J. Majak, M. Pohlak, R. Küttner, M. Eerme, K. Karjust, Artificial neural networks and genetic Algorithms in Engineering Design, in: *International Conference on Engineering Optimization –Eng Opt 2008*, Rio de Janeiro, Brazil, 2008, 01–05 June 2008.
- [11] H. Zhou, G. Sun, S. Fu, J. Liu, X. Zhou, J. Zhou, A big data mining approach of PSO-Based BP neural network for financial risk management with IoT, *IEEE Access* 7 (2019) 154035–154043 [Google Scholar][CrossRef].
- [12] X. Wu, M. Yuan, H. Yu, Soft-sensor modeling of cement raw material blending process based on fuzzy neural networks with particle swarm optimization, in: *Proceedings of the International Conference on Computational Intelligence and Natural Computing (CINCO'09)*, 2009, pp. 158–161.
- [13] J. Yaun, L. Zhong, H. Du, H. Tao, Prediction of free lime content in cement clinker based on RBF neural network, *J. Wuhan Univ. Technol. Master. Sci. Ed* 27 (2012) 187–190, 2012.
- [14] P. Zhao, Z. Zhao, Y. Chen, An improved combination modeling method for free lime content in cement clinker, in: *Proceedings of 2020 Chinese Automation Congress*, Shanghai, China, 2020, pp. 4400–4405, 6–8 November 2020.
- [15] S. Kermani, R. Behbahani, Prediction of lime saturated factor in cement raw mix using artificial neural networks, *J. Cem. Res.* 5 (2) (2015) 1–8.
- [16] Y. Zhang, Z. Li, Prediction of lime saturated factor in cement raw mix using support vector regression, *J. Build. Mater.* 2 (3) (2017) 439–446.
- [17] G. Nancy, S. Gurjot, Gaba, Harsimranjit S. Gill, A new approach for function optimization using hybrid GA-ANN algorithm, *Int. J. Eng. Res. Appl.* 2 (2) (2012) 386–389.
- [18] L.B. Adamolekun, M.A. Saliu, A.I. Lawal, I.A. Okewale, Geotechnical assessment of selected lateritic soils in southwestern Nigeria for road construction and development of artificial neural network mathematical based model for prediction of the California bearing ratio, *IJISRT* 9 (6) (2024) 1482–1491, <https://doi.org/10.38124/ijisrt/IJISRT24JUN753>.
- [19] G.M. Abdolrasol Maher, S.M. Suhail Hussain, Taha Selim Ustun, Mahidur R. Sarker, Mohammad A. Hannan, Ramizi Mohamed, Jamal Abd Ali, Saad Mekhilef, Abdalrhman Milad, Artificial neural networks based optimization techniques: a review, *Electronics* 10 (21) (2021) 2689, <https://doi.org/10.3390/electronics10212689>, 2021.
- [20] L.B. Adamolekun, M.A. Saliu, A.I. Lawal, I.A. Okewale, Development of machine learning-based standalone GUI application for predicting hydraulic conductivity and compaction parameters of lateritic soils, *Sci. Afr.* (2024), <https://doi.org/10.1016/j.sciaf.2024.e02393>, 2024–12.
- [21] CP. Dancy, J. Reidy, Review and comparison of methods to study the contribution of variables in artificial neural networks models, *Ecol. Modell* 160 (3) (2007) 249–264, [https://doi.org/10.1016/S0304-3800\(02\)00257-0](https://doi.org/10.1016/S0304-3800(02)00257-0).
- [22] E. Momeni, R. Nazir, A.D. Jahed, H. Maizir, Prediction of pile bearing capacity using a hybrid genetic algorithm-based ANN, *Measurement* 57 (2014) 122–131.
- [23] M. Monjezi, H.A. Khoshalan, M. Razifard, A neuro-genetic network for predicting uniaxial compressive strength of rocks, *J. Geotech. Geoenviron. Eng.* 30 (4) (2012) 1053–1062.
- [24] L.B. Adamolekun, M.A. Saliu, A.I. Lawal, I.A. Okewale, Investigating the competency of some soft computing techniques for modeling of lateritic soil strength based on index properties, *IJSRA* 12 (02) (2024) 47–59, <https://doi.org/10.3574/ijisra.2024.12.2.1199>.